

Murugesh Aravind

Senior Frontend Engineer | React • Micro-Frontends | TypeScript | Generative AI & Agents | AWS Certified
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Professional Summary

Senior Frontend Engineer with **8+ years** of experience building scalable, high-performance web applications using **React**, **TypeScript**, and **JavaScript**. Proven expertise as a **Tech Lead** and **Frontend Architect** in micro-frontend architecture, state management (Redux Toolkit, Zustand), and performance optimization (Core Web Vitals, code-splitting, lazy loading), backed by enterprise delivery at Cognizant and Infosys. **Fully certified in Generative AI and Large Language Model (LLM) agent development** (Google Gemini ADK, Anthropic Claude API, AI Agent Orchestration) and **AWS Certified Cloud Practitioner**, bridging the gap between robust UI architecture and production-grade agentic AI workflows. Track record of leading Angular-to-React migrations, improving load times by **40%**, achieving **85%+ test coverage**, and delivering secure, compliance-grade frontends in Agile/Scrum environments. Targeting **Staff Engineer** or senior product engineering roles in modern web and AI-powered platform teams.

Key Skills

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3, SCSS
Frameworks & Libraries: React.js, Redux Toolkit, Zustand, Context API, React Router, React Hooks, Angular
Architecture: Micro-frontends, Component-Driven Development, Single Page Applications (SPA), REST APIs
Generative AI & Agents: Google Gemini ADK, Anthropic Claude API, AI Agent Orchestration, Prompt Engineering, Context Engineering, LLM Integration, Vibe Coding (Cursor AI)
Cloud & DevOps: AWS (Cloud Practitioner), CI/CD Pipelines, Git, GitHub, Jenkins, npm, yarn, Webpack, Babel
Performance: Core Web Vitals, Code Splitting, Lazy Loading, Tree Shaking, Bundle Optimization, Chrome DevTools
Testing: Jest, React Testing Library, Playwright, Enzyme, Unit/Integration/E2E Testing, TDD
UI & Accessibility: Responsive Design, Semantic HTML, WCAG 2.1 AA, ARIA, Cross-Browser Compatibility
Methodologies: Agile, Scrum, Kanban, Code Review, System Design, Technical Mentoring, JIRA

Work Experience

Senior Frontend Engineer / Tech Lead (Senior Associate – Projects) *Bengaluru, India (Hybrid)*
Cognizant Technology Solutions *Aug 2022 – Present*

- Architected and delivered a React and TypeScript-based frontend for the **Open Account Online (OAO)** high-security banking platform, enabling compliant, high-availability customer onboarding flows for thousands of daily users
- Implemented **PII data-masking**, secure rendering patterns, and **Role-Based Access Control (RBAC)** to meet enterprise banking compliance and regulatory requirements; adopted as the internal security benchmark across engineering teams
- Optimized frontend performance using lazy loading, memoization, React Hooks, and Redux Toolkit state management, achieving a **25% improvement** in page load times and Core Web Vitals scores
- Established a comprehensive testing strategy using Jest and React Testing Library, achieving **85%+ code coverage** — adopted as the internal quality benchmark across engineering teams
- Led code reviews, enforced engineering best practices, and mentored junior frontend developers in React, TypeScript, and component-driven development

- Integrated AI-assisted development tooling (Cursor AI / GitHub Copilot) into team workflows, improving sprint velocity and code quality

Senior Frontend Engineer / Frontend Architect (Senior Associate Consultant) *Bengaluru, India*
(Remote)
Infosys Limited *Jun 2020 – Apr 2022*

- Led end-to-end Angular-to-React migration of a **Red Hat Process Automation Manager** enterprise workflow platform serving **50,000+ active users**, re-architecting the SPA for improved scalability and developer experience
- Achieved **40% reduction** in application load times and a **20-point Lighthouse score improvement** via code-splitting, lazy loading, tree-shaking, and Webpack bundle optimization — with **zero production rollbacks**
- Drove **15% growth in Daily Active Users (DAU)** within 6 months post-release through UX improvements and REST API integration optimizations
- Led **WCAG 2.1 AA** accessibility initiative across critical enterprise workflows, implementing ARIA landmarks, semantic HTML, and full keyboard navigation support
- Collaborated with backend teams on REST API design and integration, leveraging custom React Hooks and optimistic UI patterns to improve data-fetching efficiency

Software Engineer (Frontend Engineer) *Bengaluru, India (On-site)*
Amazecodes Solutions Pvt. Ltd *Jun 2018 – Mar 2020*

- Developed pixel-perfect, responsive React UIs for enterprise SaaS applications, including an HR workflow platform and financial solutions for Wipro (Project Quantum), using component-driven architecture and SCSS modules
- Built reusable, cross-browser-compatible React component libraries (Chrome, Firefox, Safari, Edge), reducing UI-related bugs by **30%** and accelerating feature delivery cycles
- Resolved critical frontend performance bottlenecks through React profiling, memoization, and code-splitting, improving application stability and user retention
- Integrated REST APIs and managed application state with Redux, delivering dynamic, data-driven UIs for B2B SaaS clients

Configuration Management Engineer *Chennai & Bengaluru, India*
Altran Technologies (Nokia Networks) & Manpower Consultancy *Dec 2014 – May 2018*

- Managed configuration and optimization of telecom infrastructure for Vodafone and Turk Telekom at Nokia Networks under Altran payroll
- Gained first web technology exposure at Target as a Digital Vendor Marketing Publisher, creating web page layouts and product content using Web Commerce Sphere
- Developed strong analytical and systems-thinking skills through telecom infrastructure work — understanding network reliability, fault tolerance, and uptime requirements that directly inform how I approach frontend resilience and production-grade software delivery today
- Completed self-directed training in JavaScript, React, HTML, and CSS, leading to a deliberate full-time transition into software engineering in 2018

Generative AI & Agent Development

Building and shipping production-grade AI-powered frontend experiences using modern agentic frameworks, including:

- **Gemini ADK Agent Orchestration:** Designed and deployed multi-agent workflows using Google Gemini Agent Development Kit (ADK) — certified via official Google enterprise program
- **Claude API Integration:** Built agentic frontend features using Anthropic Claude API, integrating context engineering principles for reliable, low-hallucination AI outputs

- **Context Engineering:** Applied structured context engineering techniques (certified) to design LLM prompt pipelines that improve AI reliability in production web applications
- **Vibe Coding / AI-Assisted Engineering:** Leveraged Cursor AI and GitHub Copilot in team workflows; competed and delivered in Cognizant's internal Vibe Code Hackathon using Cursor

Education

Bachelor of Technology – Information Technology
K.S.Rangasamy College of Technology, Tamil Nadu, India

Aug 2010 – Jun 2014
CGPA: 7.19/10

Certifications

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| • Claude Certified Architect - Foundations – Anthropic | Valid: Jun 2026 – Jun 2027 |
| • AWS Certified Cloud Practitioner – Amazon Web Services | Valid: Mar 2026 – Mar 2029 |
| • GitHub Copilot Foundations – GitHub | Valid: Jun 2025 – Jun 2028 |
| • Oracle Certified Foundations Associate – Oracle | Valid: Jun 2025 – Jun 2027 |
| • Add Agents to Gemini Enterprise (ADK) – Google | Jun 2026 |
| • Context Engineering Foundation – Cognizant | Apr 2026 |
| • AI-Assisted Development (Cursor AI / Vibe Code) – Cognizant Internal Hackathon | Nov 2025 |

Awards & Recognition

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| • Doing The Right Thing – Cognizant (Excellence in OAO Banking Platform Delivery) | 2025 |
| • Working As One – Cognizant (Successful Deliveries and Continuous Improvements) | 2023 |
| • Insta Award – Infosys (Outstanding Performance, Angular-to-React Migration) | 2021 |